

Fluorimeter PlantECG

The first Polish device for evaluation functioning "hearts" plant



Instructions stayed prepared by:

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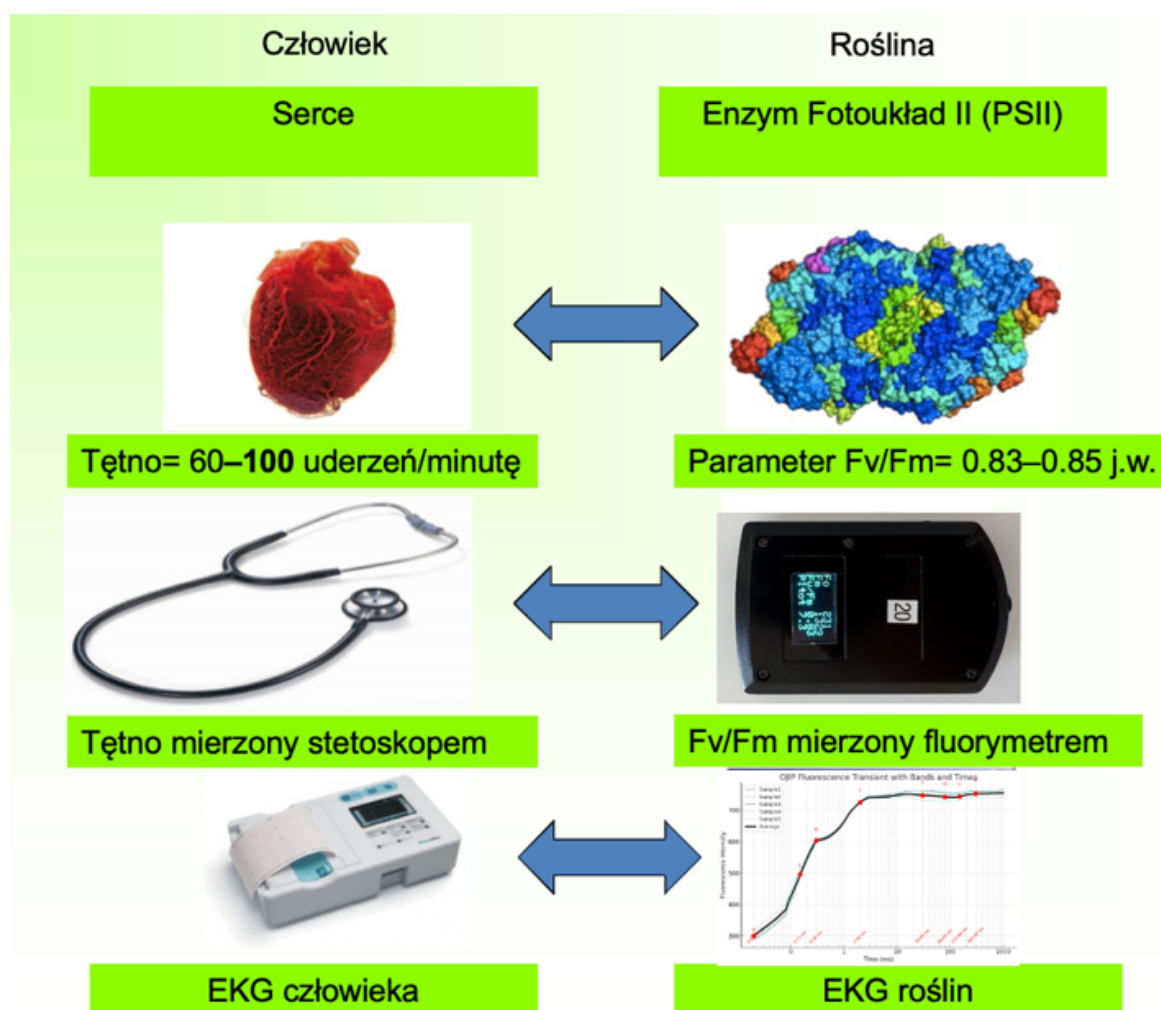
Fluorimeter PlantECG that a breakthrough tool developed by polish scientists, prof Hazema M.. Kalayi and mgr. Richard By Grodzowski, enabling early and non-invasive diagnostics condition plant. It is that the simplest, the smallest and the cheapest on world system to registration related signals process Photosynthesis (photosystem II (PSII) efficiency, located in the membranes tylacoids chloroplasts).



PlantECG it is fluorimeter, which except registration signals fluorescence chlorophyll equipped it is in sensors for measurement temperature and humidity environment and pressure atmospheric, co. ma huge meaning at interpretation obtained results. It has as well visual indicator (LED), which informs oh status physiological plants. Thanks application unique technology the device enables ultra-fast measurement (1 second), providing immediate information oh status physiologically and health plant.

Prof. Kalaji as the only one on world he conducts research on photosynthesis and analogy between human a. the plant, specifying te signals name "beating heart plants" or "electrocardiogram Plant" (ECG of plants). Prof. Kalaji he thinks, that plants they have their own language *Plantain* – and he proved, that possible it is communication z plants through process analysis photosynthesis, which is the most sensitive process life plant.

Device has been designed z they think both oh demanding professionals how and beginners users, connecting advanced technology z easy to use and interpretation results, Perfect it will check in the hands scientists z directions farmers, biological and protection environment, practitioners: farmers, the foresters, and people outside area of science and tight, how artists, architects, administrators land the greens in cities and areas protected.



Why Plant ECG?

Think how electrocardiogram (ECG) it allows you to assess the state of human health, yes Plant ECG allows you to registration "the pulse plants" – physiologically parameters life test plants. That revolutionary the device opens a new era in monitoring and protection plants, supporting agriculture the future.

Standard measurement fluorescence chlorophyll (for example The OJIP method, FV/FM, Plabs, Pitot) it is from long time applied in research physiology of plants, however adaptation this one methods to forms analogous to ECG Plants (PlantECG), how he proposed prof Kalaji, it is essential new, which are characterized by:

New diagnostic shot – instead of classical presentation results in form curves fluorescence, data interpreted they are in way analogous to recording electrocardiogram, co. gives immediately and intuitive assessment status physiological plants.

Unique technological solution – developed device (fluorimeter PlantECG) links simplicity, miniaturization and multi-functionality (for example built-in sensors temperature, humidity, pressure) z possibility ultra-fast measurement (1 second).

Practical application – by adapting to forms "ECG plants", results they are more understandable no, no. only for scientists, but also for farmers, gardeners or the decision-makers, co. it makes it easy transfer knowledge and implementation in practice agricultural.

Originality and priority – to this one time of no one in literature scientific neither in technology commercially no, no. introduced concept "ECG plants" in this form.

IN relationship z that you can go ahead classification that as innovative technology in field of diagnosis plant and potential subject of patent protection.

Definition of new technology PlantECG:

PlantECG (Electrocardiogram *of plants*) that innovative technology diagnosis developed by prof. Hazema M.. By Kalayi, the adaptation of measurements fluorescence chlorophyll to forms analogous to recording electrocardiogram man. Te technology ta enables ultra-fast, non-invasive assessment condition physiological plants and early detection symptoms stress biotic and abiotic.

Elements of novelty and innovation:

New concept measurement – transformation of data fluorescent in form "signal pulsing" plants, the responder rhythm operation of the system photosynthetic.

First device this type – fluorimeter PlantECG it is the smallest and the simplest device on the world, which integrates sensors environmental (temperature, humidity, pressure) and system visually status indication Physiological plant (LED in colors green, yellow, red) for immediate interpretation results.



New method interpretation data – simplified save reminder ECG it allows you to intuitive analysis results, understood both for scientists, how and practitioners agriculture or gardening.

Application practical – technology supports agriculture precise, sustainable management crops and systems early warning oh stress environmental.

Meaning and potential implementation:

Scientific – introduces a new paradigm in research on signaling and fault finding photosynthetic.

Industrial – allows you to cheap, fast and universal health check plant in farms agricultural, greenhouses and laboratories.

Socio-economic – makes it easier transfer knowledge to sector agri-food, increasing resistance systems production food on climatic stress.

The most important thing functions

- ! Non-invasive and non-contact measurement – without necessity damage plants.
- ! Ultra fast fault finding – result in time the real, come on in 1 second.
- ! Versatility applications – w ecological research (land and water), agriculture, gardening and forestry.
- ! Assessment condition physiologically and health – precise analysis functioning The photosystem II.
- ! Early detection stress u plants (abiotic and (biotic) – includes stress of drought, salinity, shortages components food, impurities environmental, and also factors biotic, such as infections pathogens, fungal infections, diseases viral and bacterial and others forms biological pressure affecting on condition and functioning of plants.
- ! Wireless transmission data to computer or devices mobile.
- ! Measurement parameters environment at everyone use – temperature, pressure atmospheric, relative humidity.
- ! Alphanumeric value display measured parameters.
- ! Indicator visual health status plant – colored LED to monitoring condition plants (color green – healthy plant, color gold- orange – plant in moderate stress, the color red – plant in the deep stress).

Specifications technical

- Type measurement: non-invasive, tactile z possibility measurement without adaptation plant in darkness.
- Measurement time (fast induction fluorescence chlorophyll (OJIP): 1 second
- Light source by the exciter: Blue LED (430 nm)
- Intensity lights by the excerpt (illumination): about 4000 $\mu\text{mol photon/m}^2/\text{sec}$.